

Network Traffic Analysis Using Wireshark

Introduction

This project focuses on capturing and analyzing network traffic using Wireshark. The goal is to understand network communication and identify common protocols used in everyday internet activity.

Objective

- Capture network packets.
- Analyze DNS traffic.
- Analyze TCP connections.
- Examine TLS encrypted traffic.
- Observe ICMP packets.

Tools Used

- Wireshark
- Windows 11
- Internet Connection

Methodology

1. Installed Wireshark.
2. Captured network traffic through the active Wi-Fi adapter.
3. Generated traffic by browsing websites.
4. Applied filters for DNS, TCP, TLS, and ICMP.
5. Saved screenshots and packet capture files.
6. Analyzed protocol behavior.

Results

DNS Analysis

DNS packets were observed translating domain names into IP addresses.

TCP Analysis

TCP packets demonstrated reliable communication between hosts.

TLS Analysis

TLS packets showed encrypted communication protecting user data.

ICMP Analysis

ICMP packets were captured using ping requests to test connectivity.

Conclusion

This project provided practical experience in packet capture and network traffic analysis. The knowledge gained is relevant to SOC Analyst roles, where monitoring and investigating network activity are essential responsibilities.